END-USER DOCUMENTATION

BROKEN FM

Real-time video frequency modulation.
Companion UI and OFX parameter reference.
Every exposed parameter, explained.

VERSION 0.1.0 / EN / 82 PARAMETERS / 56 FACTORY PRESETS

About BROKEN FM

BROKEN FM turns image structure into a generated electronic signal. It is not a displacement, crop, or scaling effect: source pixels are analyzed as modulation while the carrier is built in output pixel space.

Photosensitivity warning. BROKEN FM can generate flashing, flickering, and high-contrast moving patterns. The application shows this warning before the renderer or test pattern starts. Its checkbox suppresses future warnings on the same device; clearing local application storage restores it.

Native OFX v0.1 capability note. PM/FM, procedural carriers, LFO routing, instability, dropout, color, and output shaping run natively. Audio carrier/analysis, imported custom wavetable payloads, feedback history, and phosphor history remain parity gates. The Companion session disables unsupported choices where possible; shared Resolve enum and matrix controls may still expose combinations that the native renderer rejects instead of approximating.

Quick start

1. Start from a preset

Choose a factory preset in the header and use previous/next to explore the range.

2. Choose the source

After the warning is acknowledged, Test pattern is active by default. Load video enables Play; Test pattern pauses it.

3. Shape modulation

Choose source analysis, gain/bias, threshold, and overall Modulation Gain.

4. Choose PM or FM

PM changes phase locally. FM accumulates frequency change along Scan Angle.

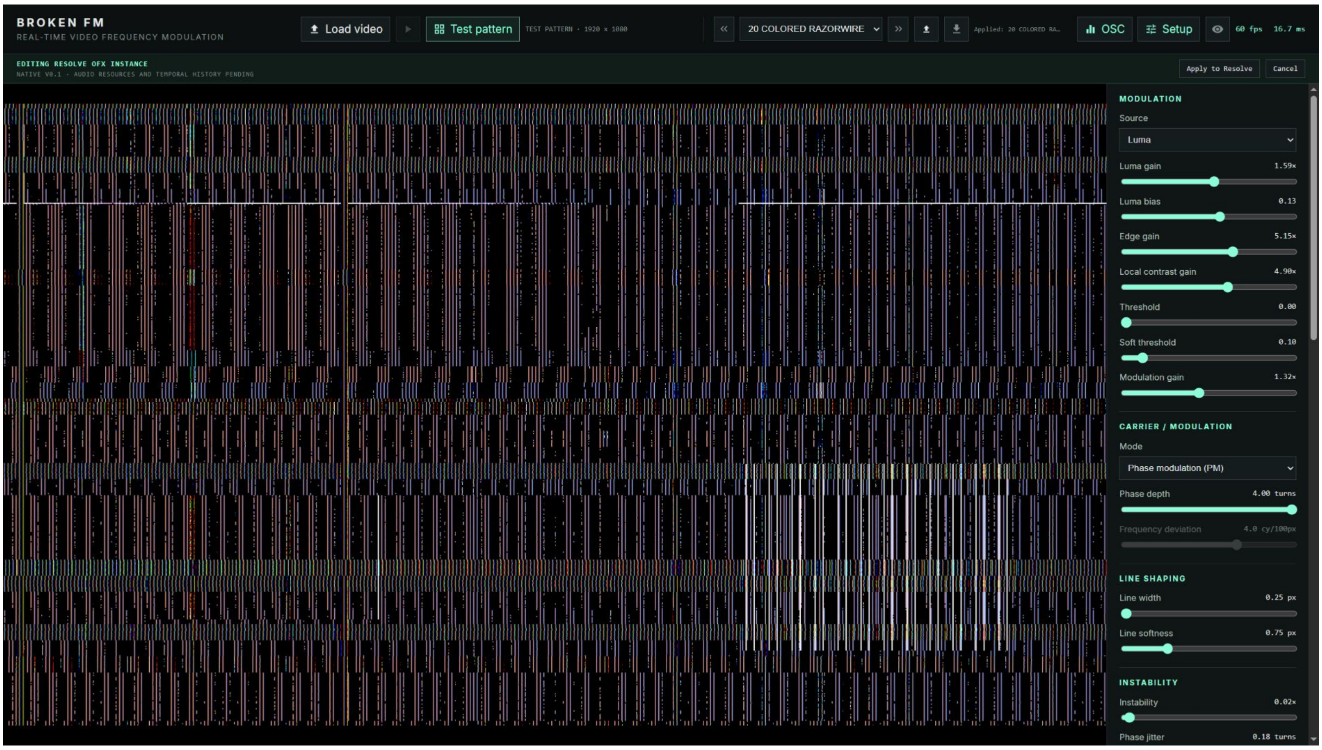
5. Add controlled motion

Use Instability or route LFO/audio analysis sources in OSC.

6. Save or apply

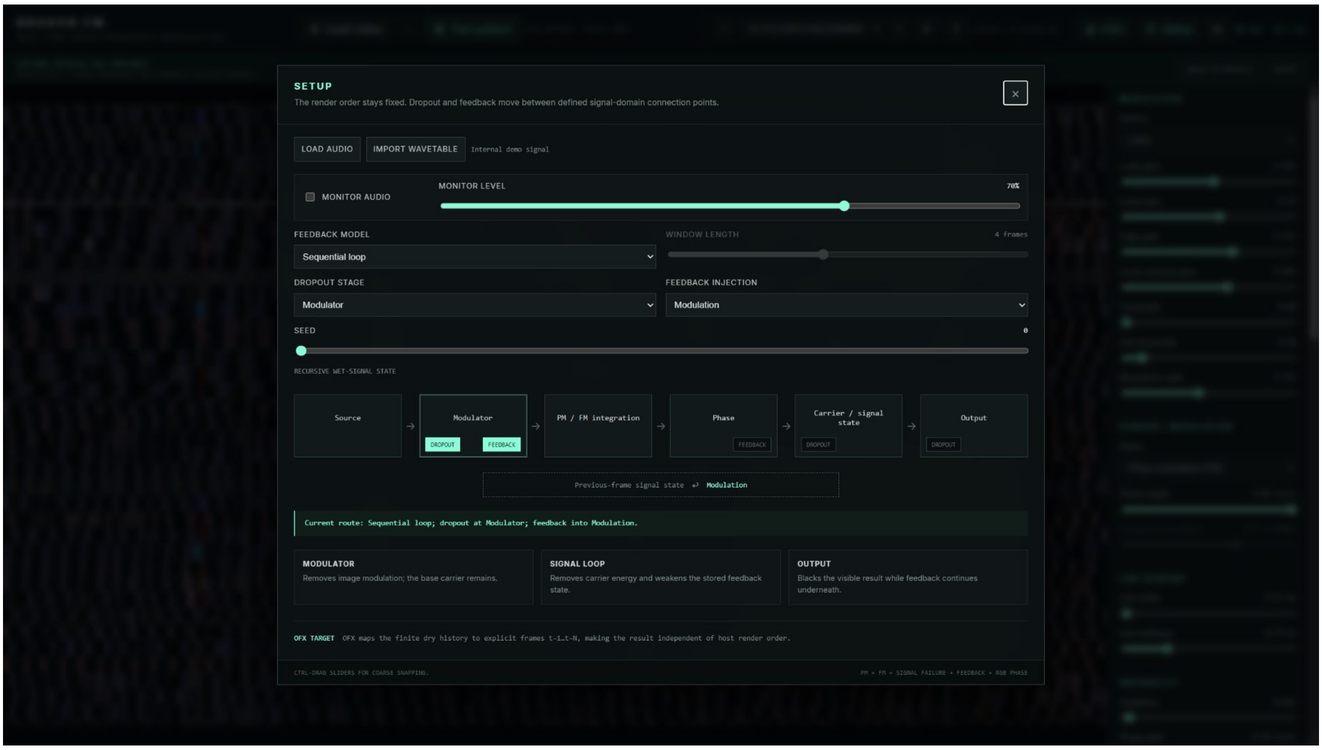
Save JSON stores the effect state. In a Resolve session, Apply returns it to OFX.

Interface tour

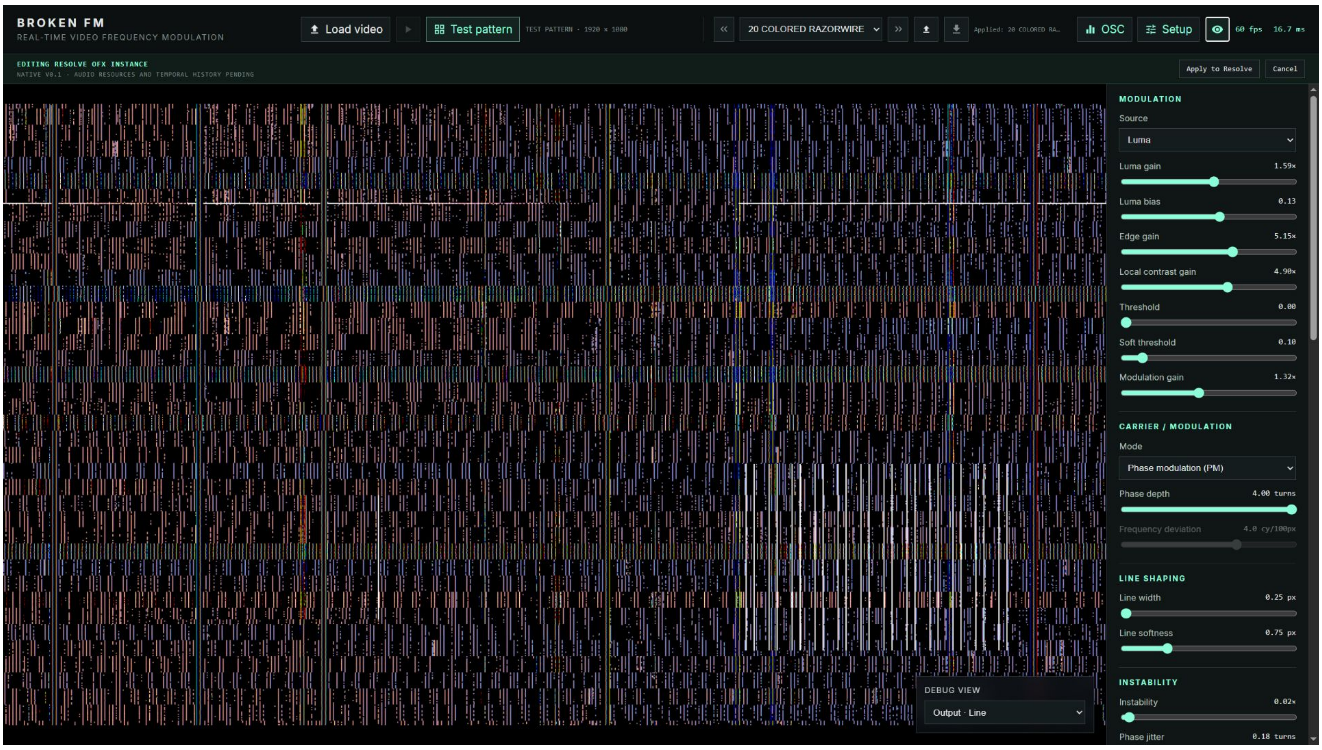


Main interface at 1920 x 1080 using 20 COLORED RAZORWIRE.

OSC/LFO modal with carrier preview, two LFOs, and modulation matrix.



Setup modal showing active dropout and feedback connection points.



Optional Debug View overlay in the lower-right of the video preview.

Controls and signal path

Normal slider dragging is fine-grained. Hold Ctrl while dragging for meaningful coarse snapping. Gray controls are inactive in the current mode. Matrix-modulated destinations show base -> effective while their value changes.

The fixed path is source analysis -> threshold/gain -> PM or scan-integrated FM -> carrier rendering -> color/output. Setup moves Dropout and Feedback between defined connection points. Phosphor persistence is display-only and never feeds the signal loop.

Complete parameter reference

Modulation source

Mod Source

`modSource`

Values luma, inverted-luma, edges, local-contrast, luma-edges
Default luma

Chooses the image feature that drives modulation: luminance, inverted luminance, edges, local contrast, or a luminance-plus-edge blend.

Luma Gain

`lumaGain`

Values 0 to 3
Default 1

Multiplies source luminance before centering it around the neutral midpoint. Higher values make tonal differences drive the carrier more strongly.

Luma Bias

`lumaBias`

Values -1 to 1
Default 0

Offsets luminance before it becomes a modulation signal. Use it to choose which tonal region sits near the neutral carrier state.

Edge Gain

`edgeGain`

Values 0 to 8
Default 2.5

Sets the strength of the extracted edge signal. It is most obvious with Edges or Luma + edges as Mod Source.

Local Contrast Gain

`localContrastGain`

Values 0 to 8
Default 4

Sets the strength of local brightness differences. Higher values emphasize texture and small regional changes rather than overall exposure.

Mod Threshold

`modThreshold`

Values 0 to 1
Default 0

Suppresses modulation below the selected level. A value of 0 is an exact bypass; higher values isolate progressively stronger source features.

Mod Softness

`modSoftness`

Values 0 to 1
Default 0.1

Controls the transition width around Mod Threshold. At 0 the threshold is hard; higher values create a smoother gate. It has no visible effect when Threshold is 0.

Modulation Gain

`modulationGain`

Values 0 to 3
Default 1

Master gain for the conditioned image modulation before PM or FM. At 0 the source contributes no information to the generated signal.

Carrier and PM/FM

Mode

`mode`

Values pm, fm
Default pm

Selects the signal model. PM adds the source value directly to carrier phase. FM integrates source-driven frequency changes along the selected scan direction, producing folds and accumulated bends.

Frequency

`frequency`

Values 0.5 to 30 cy/100px
Default 8 cy/100px

Base carrier density in cycles per 100 output pixels along Scan Angle. Higher values create more closely spaced lines or waveform cycles.

Carrier Phase

`carrierPhase`

Values -1 to 1 turns
Default 0 turns

Offsets the carrier by whole turns without moving source pixels. Animate it to make the generated signal travel through the image.

Carrier Shape

`carrierShape`

Values sine, triangle, saw, square, noise, wavetable, audio
Default sine

Selects Sine, Triangle, Saw, Square, Noise, Wavetable, or Audio as the carrier. Audio requires a loaded audio source in the Companion preview.

Signal Render

`signalRender`

Values line, waveform
Default line

Line draws narrow contours at repeating carrier phases. Waveform displays the continuous bipolar signal as image intensity.

Pwm

`pwm`

Values 0.05 to 0.95
Default 0.5

Changes duty cycle for Square and skew/symmetry for Triangle and Saw-derived shapes. 0.5 is centered and symmetrical.

Noise Detail

`noiseDetail`

Values 0.25 to 8
Default 2

Controls spatial detail for the Noise carrier. Low values make broad variations; high values make finer, faster-changing structure.

Wavetable

`wavetable`

Values harmonic, folded, formant, custom
Default harmonic

Selects the internal Harmonic, Folded, or Formant table, or the session-only imported Custom table. Custom data is not embedded in preset JSON.

Phase Depth

`phaseDepth`

Values 0 to 4 turns
Default 0.8 turns

PM depth in full turns. It controls how far image modulation bends carrier phase; 0 removes PM influence even if Modulation Gain is non-zero.

Frequency Deviation

`frequencyDeviation`

Values -12 to 12 cy/100px
Default 4 cy/100px

FM deviation in cycles per 100 pixels at full-scale modulation. Positive and negative values reverse the direction of accumulated frequency bending. Active only in FM mode.

Scan Angle

`scanAngle`

Values -180 to 180 degrees
Default 0 degrees

Direction in which the carrier and FM integral travel, from -180 degrees to 180 degrees. It rotates the signal coordinate, not the source image.

Line shaping

Line Width

`lineWidth`

Values 0.25 to 5 px
Default 1.2 px

Approximate on-screen thickness of Line render contours in pixels.

Line Softness

`lineSoftness`

Values 0 to 3 px
Default 0.75 px

Antialiasing and edge softness around Line render contours. 0 is crisp; larger values create broader, softer edges.

Instability

Instability

`instability`

Values 0 to 1
Default 0

Master amount for drift and jitter contributions. At 0 the clean PM/FM path is restored even when the individual instability controls have non-zero setup values.

Phase Jitter

`phaseJitter`

Values 0 to 1 turns
Default 0.18 turns

Adds smooth, deterministic disturbance directly to carrier phase. It roughens and wriggles the signal without adding a visible noise overlay.

Signal Noise

`signalNoise`

Values 0 to 1
Default 0

Injects deterministic broadband disturbance into signal phase. It produces RF-like breakup while remaining part of the carrier calculation.

Frequency Drift

`frequencyDrift`

Values 0 to 8 cy/100px
Default 1.5 cy/100px

Maximum slow variation of carrier frequency in cycles per 100 pixels. Its motion rate is controlled by Drift Rate.

Horizontal Drift

`horizontalDrift`

Values 0 to 120 px
Default 0 px

Moves generated signal coordinates horizontally in pixel space. The source image is not resampled or translated.

Vertical Drift

`verticalDrift`

Values 0 to 120 px
Default 0 px

Moves generated signal coordinates vertically in pixel space. Combine it with Horizontal Drift for wandering two-axis motion.

Drift Rate
`driftRate`

Values 0.01 to 5 Hz
Default 0.22 Hz

Speed in hertz of frequency and coordinate drift fields. Low values wander slowly; high values hunt rapidly.

Time Jitter
`timeJitter`

Values 0 to 30 frames
Default 0 frames

Perturbs the internal signal clock by an amount expressed in frames. It affects procedural motion and failure timing, but does not select a different source-video frame.

Line Jitter
`lineJitter`

Values 0 to 40 px
Default 5 px

Maximum displacement of scan-line bands in pixels. This is the amplitude of horizontal-sync-like line movement.

Line Jitter Scale
`lineJitterScale`

Values 1 to 128 px
Default 2 px

Width of the bands affected by Line Jitter. Small values produce fine buzzing lines; large values shift broader blocks together.

Line Jitter Speed
`lineJitterSpeed`

Values 0 to 30 Hz
Default 7 Hz

Temporal speed in hertz of the Line Jitter pattern. 0 freezes the pattern for a given frame state.

Signal failure

Dropout
`dropout`

Values 0 to 1
Default 0

Probability/intensity of deterministic missing-signal bands. The affected signal-domain location is selected by Dropout Stage.

Dropout Angle
`dropoutAngle`

Values -180 to 180 degrees
Default 0 degrees

Orientation of dropout bands, independent of Scan Angle. Use a different angle to cross-cut the carrier structure.

Line Tear
`lineTear`

Values 0 to 120 px
Default 0 px

Maximum discontinuous jump of the scan coordinate in pixels. It tears carrier phase and FM lookup together instead of shifting the finished image.

Dropout Stage
`dropoutStage`

Values modulator, signal-loop, output
Default modulator

Chooses where dropout enters the route: Modulator removes source modulation, Signal loop removes carrier energy and weakens stored feedback, and Output blacks only the visible result.

Feedback

Feedback Amount

`feedbackAmount`

Values 0 to 1
Default 0

Coupling strength of the stored signal returned to the current frame. 0 disables feedback.

Feedback Decay

`feedbackDecay`

Values 0 to 0.99
Default 0.9

Retention applied to older signal state. Values near 1 create longer, more persistent echoes.

Feedback Displacement

`feedbackDisplacement`

Values 0 to 64 px/frame
Default 0 px/frame

Moves stored signal by pixels per age step before reinjection. It creates trails, conveyors, and drifting memory rather than a simple overlay.

Feedback Displacement Angle

`feedbackDisplacementAngle`

Values -180 to 180 degrees
Default 0 degrees

Direction of feedback displacement from -180 degrees to 180 degrees.

Feedback Phase

`feedbackPhase`

Values -1 to 1 turns
Default 0 turns

Rotates the stored waveform and quadrature pair in turns before reinjection. This changes echo phase and polarity without disguising the operation as gain.

Feedback Injection

`feedbackInjection`

Values phase, modulation
Default phase

Phase injects stored state after PM/FM integration. Modulation returns it before integration, so FM can accumulate and reshape the feedback.

Feedback Model

`feedbackModel`

Values sequential, finite
Default sequential

Sequential loop recursively uses the previous wet state and depends on linear playback. Finite window combines a bounded set of previous dry frames for a deterministic history length.

Feedback Window

`feedbackWindow`

Values 1 to 8
Default 4

Number of previous dry frames, 1 through 8, used by Finite window. It is disabled for Sequential loop.

Color

Color Mode

`colorMode`

Values mono, input-color, rgb-phase
Default mono

Mono outputs one signal in all channels. Input color applies normalized source chroma to signal energy. RGB phase evaluates red, green, and blue at different carrier phases.

Rgb Phase Offset

`rgbPhaseOffset`

Values 0 to 0.5 turns
Default 0.035 turns

Phase separation in turns between RGB channels. Active only in RGB phase mode; larger offsets create stronger color splitting.

Output

Black Level

`blackLevel`

Values -0.5 to 0.5
Default 0

Adds or removes output pedestal before contrast, gamma, and brightness shaping. Negative values crush low signal; positive values lift black.

Signal Gamma

`signalGamma`

Values 0.25 to 4
Default 1

Nonlinear response of the generated signal. Values below 1 brighten mid-levels; values above 1 darken them.

Brightness

`brightness`

Values 0 to 3
Default 1

Final output gain after black level, contrast, and gamma. At 0 the visible output is black.

Contrast

`contrast`

Values 0.25 to 4
Default 1

Expands or compresses output around mid-gray before gamma. 1 is neutral.

Phosphor Persistence

`phosphorPersistence`

Values 0 to 3000 ms
Default 0 ms

Peak-hold afterglow half-life in milliseconds. It stores bright RGB emission while decaying older light; it never feeds back into the carrier or modulation path.

Audio analysis

Audio Attack

`audioAttack`

Values 0 to 500 ms
Default 25 ms

Rise time in milliseconds for Level, Low, Mid, High, and Transient analysis. Short attack follows hits quickly; long attack smooths their onset.

Audio Release

`audioRelease`

Values 0 to 2000 ms
Default 180 ms

Fall time in milliseconds for audio analysis. Long release holds energy after a sound; short release makes modulation drop quickly.

Low-frequency oscillators

Lfo1 Shape

`lfo1Shape`

Values sine, triangle, saw, square, sample-hold, smooth-noise
Default sine

Waveform generated by LFO 1: Sine, Triangle, Saw, Square, stepped Sample & hold, or interpolated Smooth noise.

Lfo1 Rate Mode

`lfo1RateMode`

Values hz, bpm
Default hz

Chooses free-running Hz or tempo-based BPM timing for LFO 1. Only the controls for the selected rate mode affect its speed.

Lfo1 Sync

`lfo1Sync`

Values timeline, transport
Default timeline

Timeline evaluates LFO 1 from absolute host time. Transport subtracts the most recent retrigger origin so playback/retrigger can restart its cycle.

Lfo1 Rate Hz

`lfo1RateHz`

Values 0.01 to 20 Hz
Default 0.25 Hz

Free-running speed of LFO 1 in cycles per second. Active when Rate mode is Hz.

Lfo1 Bpm

`lfo1Bpm`

Values 20 to 300 BPM
Default 120 BPM

Tempo used by LFO 1 for musical timing. Active when Rate mode is BPM.

Lfo1 Division

`lfo1Division`

Values 2-bars, 1-bar, 1-2, 1-4, 1-8, 1-16, 1-32
Default 2-bars

Musical cycle length for LFO 1, from two bars to 1/32. In 4/4, 1/4 produces one cycle per beat.

Lfo1 Phase
`lfo1Phase`

Values 0 to 1 turns
Default 0 turns

Starting phase of LFO 1 in turns. 0.25 is a quarter-cycle offset and 0.5 is half a cycle.

Lfo1 Polarity
`lfo1Polarity`

Values bipolar, unipolar
Default bipolar

Bipolar maps LFO 1 to -1...+1 for motion around the base value. Unipolar maps it to 0...+1 for one-direction modulation.

Lfo2 Shape
`lfo2Shape`

Values sine, triangle, saw, square, sample-hold, smooth-noise
Default sine

Waveform generated by LFO 2: Sine, Triangle, Saw, Square, stepped Sample & hold, or interpolated Smooth noise.

Lfo2 Rate Mode
`lfo2RateMode`

Values hz, bpm
Default hz

Chooses free-running Hz or tempo-based BPM timing for LFO 2. Only the controls for the selected rate mode affect its speed.

Lfo2 Sync
`lfo2Sync`

Values timeline, transport
Default timeline

Timeline evaluates LFO 2 from absolute host time. Transport subtracts the most recent retrigger origin so playback/retrigger can restart its cycle.

Lfo2 Rate Hz
`lfo2RateHz`

Values 0.01 to 20 Hz
Default 0.1 Hz

Free-running speed of LFO 2 in cycles per second. Active when Rate mode is Hz.

Lfo2 Bpm
`lfo2Bpm`

Values 20 to 300 BPM
Default 120 BPM

Tempo used by LFO 2 for musical timing. Active when Rate mode is BPM.

Lfo2 Division

`lfo2Division`

Values 2-bars, 1-bar, 1-2, 1-4, 1-8, 1-16, 1-32
Default 2-bars

Musical cycle length for LFO 2, from two bars to 1/32. In 4/4, 1/4 produces one cycle per beat.

Lfo2 Phase

`lfo2Phase`

Values 0 to 1 turns
Default 0.25 turns

Starting phase of LFO 2 in turns. 0.25 is a quarter-cycle offset and 0.5 is half a cycle.

Lfo2 Polarity

`lfo2Polarity`

Values bipolar, unipolar
Default bipolar

Bipolar maps LFO 2 to -1...+1 for motion around the base value. Unipolar maps it to 0...+1 for one-direction modulation.

Modulation matrix

Audio Source1

`audioSource1`

Values off, level, low, mid, high, transient, lfo-1, lfo-2
Default off

Source for modulation slot 1: Off, audio Level/Low/Mid/High/Transient analysis, LFO 1, or LFO 2. Audio sources require audio in the Companion preview.

Audio Destination1

`audioDestination1`

Values off, frequency, pwm, phase, phase-depth, feedback, scan-angle, dropout, phase-jitter, frequency-drift, line-jitter, feedback-displacement, feedback-displacement-angle, horizontal-drift, vertical-drift, time-jitter, phosphor-persistence, signal-noise, line-jitter-scale, line-jitter-speed, feedback-phase
Default off

Parameter controlled by modulation slot 1. Off disconnects the slot. A destination can receive multiple slots, which are summed before clamping.

Audio Amount1

`audioAmount1`

Values -1 to 1
Default 0

Signed depth of modulation slot 1. Positive values follow the source; negative values invert it. 0 disables the slot without changing its source/destination choices.

Audio Source2

`audioSource2`

Values off, level, low, mid, high, transient, lfo-1, lfo-2
Default off

Source for modulation slot 2: Off, audio Level/Low/Mid/High/Transient analysis, LFO 1, or LFO 2. Audio sources require audio in the Companion preview.

Audio Destination2

`audioDestination2`

Values off, frequency, pwm, phase, phase-depth, feedback, scan-angle, dropout, phase-jitter, frequency-drift, line-jitter, feedback-displacement, feedback-displacement-angle, horizontal-drift, vertical-drift, time-jitter, phosphor-persistence, signal-noise, line-jitter-scale, line-jitter-speed, feedback-phase
Default off

Parameter controlled by modulation slot 2. Off disconnects the slot. A destination can receive multiple slots, which are summed before clamping.

Audio Amount2

`audioAmount2`

Values -1 to 1
Default 0

Signed depth of modulation slot 2. Positive values follow the source; negative values invert it. 0 disables the slot without changing its source/destination choices.

Audio Source3

`audioSource3`

Values off, level, low, mid, high, transient, lfo-1, lfo-2
Default off

Source for modulation slot 3: Off, audio Level/Low/Mid/High/Transient analysis, LFO 1, or LFO 2. Audio sources require audio in the Companion preview.

Audio Destination3

`audioDestination3`

Values off, frequency, pwm, phase, phase-depth, feedback, scan-angle, dropout, phase-jitter, frequency-drift, line-jitter, feedback-displacement, feedback-displacement-angle, horizontal-drift, vertical-drift, time-jitter, phosphor-persistence, signal-noise, line-jitter-scale, line-jitter-speed, feedback-phase
Default off

Parameter controlled by modulation slot 3. Off disconnects the slot. A destination can receive multiple slots, which are summed before clamping.

Audio Amount3

`audioAmount3`

Values -1 to 1
Default 0

Signed depth of modulation slot 3. Positive values follow the source; negative values invert it. 0 disables the slot without changing its source/destination choices.

Audio Source4

`audioSource4`

Values off, level, low, mid, high, transient, lfo-1, lfo-2
Default off

Source for modulation slot 4: Off, audio Level/Low/Mid/High/Transient analysis, LFO 1, or LFO 2. Audio sources require audio in the Companion preview.

Audio Destination4

`audioDestination4`

Values off, frequency, pwm, phase, phase-depth, feedback, scan-angle, dropout, phase-jitter, frequency-drift, line-jitter, feedback-displacement, feedback-displacement-angle, horizontal-drift, vertical-drift, time-jitter, phosphor-persistence, signal-noise, line-jitter-scale, line-jitter-speed, feedback-phase
Default off

Parameter controlled by modulation slot 4. Off disconnects the slot. A destination can receive multiple slots, which are summed before clamping.

Audio Amount4

`audioAmount4`

Values -1 to 1
Default 0

Signed depth of modulation slot 4. Positive values follow the source; negative values invert it. 0 disables the slot without changing its source/destination choices.

Setup

Seed

seed

Values 0 to 9999
Default 0

Changes all deterministic random fields used by noise, jitter, dropout, and related modulation. The same seed, frame, input, and parameters reproduce the same result.

Operational details

Presets

Preset JSON stores all 82 effect parameters, but no source files, imported samples, monitor state, playback position, Debug View, retrigger origin, or temporal buffers. Loading a preset resets temporal history and retriggers Transport LFOs.

LFO timing

Timeline sync uses absolute host time. Transport sync uses a session retrigger origin. In 4/4, 1/4 is one cycle per beat, 1/8 is two cycles per beat, and 1 bar is one cycle per four beats.

Current OFX limits

Audio carrier/analysis, imported custom wavetable payloads, temporal feedback history, and phosphor history are explicit parity gates in OFX 0.1.0. The Companion session disables unsupported choices where possible; shared Resolve enum and matrix controls may still expose combinations that the native renderer rejects instead of approximating. Time Jitter changes internal signal time and is not source-video time remapping.